Instructor's Note:

Not part of the class project, and not extra credit. Three of the burn models in bitwhiz, the course's open-source hydrocode, need the pressure of the first shock to reach each cell, and burn only after it arrives. A hydrocode smears every shock over a few cells, so which state is "the shock"? Menikoff and Shaw's answer uses nothing but the cell's own history and the Hugoniot jump condition for energy [1]. This sheet gives it, with the rules bitwhiz adds to make it work on a mesh.

These equations build upon U5. Rankine-Hugoniot Analysis and the Class Project: Hydrocode sheets (Method AV and Method RS)

What the Lead Shock Adds to Your Hydrocode:

- Two numbers per cell, t_S, the time the lead shock arrives, and P_S, its pressure: found from the cell's own history, then frozen
- A gate, a burn model that uses P_S burns nothing before t_S ThermalCascade, FlashFire, StochasticIgnite
- Frozen means frozen: a second shock does not change P_S; it acts only through the cell's own P and T (one route to shock desensitization)
- Carried from step to step, five numbers per cell: $\mathcal{H}$ at the last step, its largest value, a step flag, P_S, t_S
- Notation, Hugoniot function $\mathcal{H}$ (Menikoff and Shaw's h; not U3's areal mass coordinate h); the state ahead of the shock $(\ )_0$; artificial viscosity q (Method AV)
- ***One check per cell per cycle, after the pressures: the hydro itself is unchanged***

The Hugoniot Function [1]:

- Energy jump, $\boldsymbol{e_H - e_0 \;=\; \frac{1}{2}\,(P_H + P_0)\,(\mathrm{V_0} - \mathrm{V}_H)}$ U5, any shock from the state $\boldsymbol{(\;)_0}$
- Hugoniot function, $\boldsymbol{\mathcal{H}\,(\mathrm{V}, e) \;=\; e - e_0 - \frac{1}{2}\,(\mathrm{V_0} - \mathrm{V})\,[P\,(\mathrm{V}, e) + P_0]}$ zero on the Hugoniot
- Erratum, Menikoff and Shaw print $\boldsymbol{P - P_0}$ in their Eq. (10); the jump condition has $\boldsymbol{P + P_0}$, and at $\boldsymbol{P_0 = 0}$ the two agree
- At rest, every cell starts at $\boldsymbol{\rho_0}$ with $\boldsymbol{e_0 = 0}$, $\boldsymbol{P_0 = 0}$: $\boldsymbol{\mathcal{H} \;=\; e - \frac{1}{2}\,(\mathrm{V_0} - \mathrm{V})\,P}$
- Each cell, after each step, $\boldsymbol{\mathcal{H}^{n+1} = \mathcal{H}\left(\mathrm{V}^{n+1}, e^{n+1}\right)}$ their Eq. (11): a history, cell by cell
- Inside a steady smeared shock, $\boldsymbol{\sigma = P + q}$ is linear in V (the Rayleigh line), and steady energy gives $\boldsymbol{e - e_0 = \frac{1}{2}\,(\sigma + P_0)\,(\mathrm{V_0} - \mathrm{V})}$, so

  . $\boldsymbol{\mathcal{H} \;=\; \frac{1}{2}\,q\,(\mathrm{V_0} - \mathrm{V}) \;\geq\; 0}$ it rises through the shock and returns to zero behind it
- Smooth, isentropic compression from rest, $\boldsymbol{\mathcal{H} \leq 0}$: the chord from $\boldsymbol{(\mathrm{V_0}, 0)}$ to $\boldsymbol{(\mathrm{V}, P)}$ lies above the convex isentrope beneath it
- ***From rest, a cell's $\mathcal{H}$ first turns positive in a shock: the lead shock is where it first rises and falls back*** (a release behind a shock keeps $\boldsymbol{\mathcal{H} > 0}$: the shock's entropy stays)

References:

[1] R. Menikoff and M. S. Shaw, "Reactive burn models and ignition & growth concept," *EPJ Web of Conferences* **10**, 00003 (2010), doi:10.1051/epjconf/20101000003.